Last Time

Summary of Sampling Methods

Suppose we select k objects from a set of n distinguishable objects.

Sampling Method	Number of Outcomes
With replacement, order matters	n^k
Without replacement, order matters	$(n)_k = \frac{n!}{(n-k)!}$
Without replacement, order does not matter	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Example. A standard deck contains 52 cards, of which 4 are queens. Suppose 7 cards are selected uniformly at random without replacement, and the order of selection is not recorded. What is the probability that you have **exactly one, two, three, or four queens**?

The sample space has $\binom{52}{7}$ outcomes.

Event $A = \{$ 7 cards with exactly one queen $\}$

$$\binom{4}{1}\cdot\binom{52-4}{6} = \binom{4}{1}\cdot\binom{48}{6}$$

select one queen from the 4 queens

$$\therefore P(A) = \frac{|A|}{|\Omega|} = \frac{\binom{4}{1}\binom{48}{6}}{\binom{52}{7}} \approx 0.3669$$

DISCRETE RANDOM VARIABLES

Example. Suppose a fair coin is tossed twice. The sample space is

$$\Omega = \{HH, HT, TH, TT\}.$$

Suppose we are interested in the **number of heads** observed.

$X: \Omega \to \mathbb{R}$

$\omega \to X(\omega)$

Define X by

$$X(HH) = 2,$$
$$X(HT) = 1,$$
$$X(TH) = 1,$$
$$X(TT) = 0.$$

Thus,

$$X : \Omega \longrightarrow \{0, 1, 2\}.$$

For example,

$$X(HT) = 1.$$

The outcomes of a random experiment do not have to be numbers. The function X assigns a numerical value to each outcome, and it allows us to formulate questions about a random experiment ***numerically***.

In this example,

$$\{X = 1\} = \{HT, TH\}.$$

Therefore,

$$P(X = 1) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}.$$

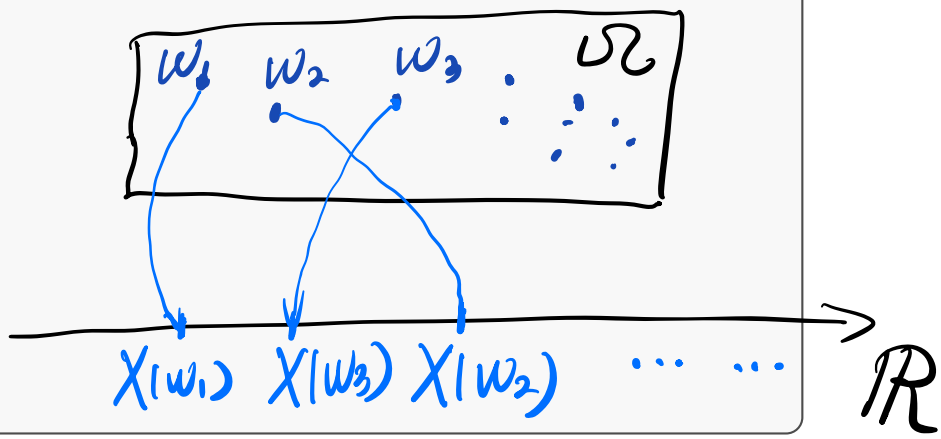
Random Variables

Definition

A **random variable** X is a function that assigns a real number to each outcome ω in the sample space Ω:

$X: \Omega \to \mathbb{R}$

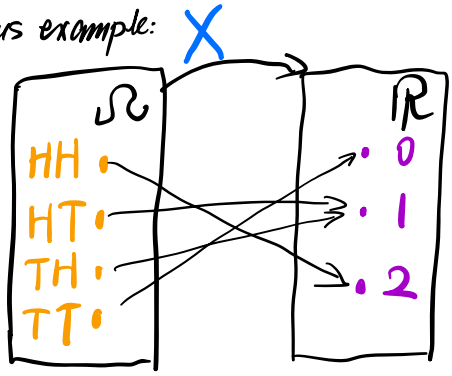
$\omega \mapsto X(\omega)$



Thus, for every outcome $\omega \in \Omega$, the random variable assigns a numerical value $X(\omega)$.

It is called "variable" because its numerical value varies depending on the random outcome of the experiment.

In the previous example: X



Notation:

- $\{X = x\}$ denotes the event $\{\omega \in \Omega \mid X(\omega) = x\}$ where $x \in \mathbb{R}$.
- $P(X = x)$ represents the probability of the event $\{X = x\}$.

$\{X=1\} = \{HT, TH\}$

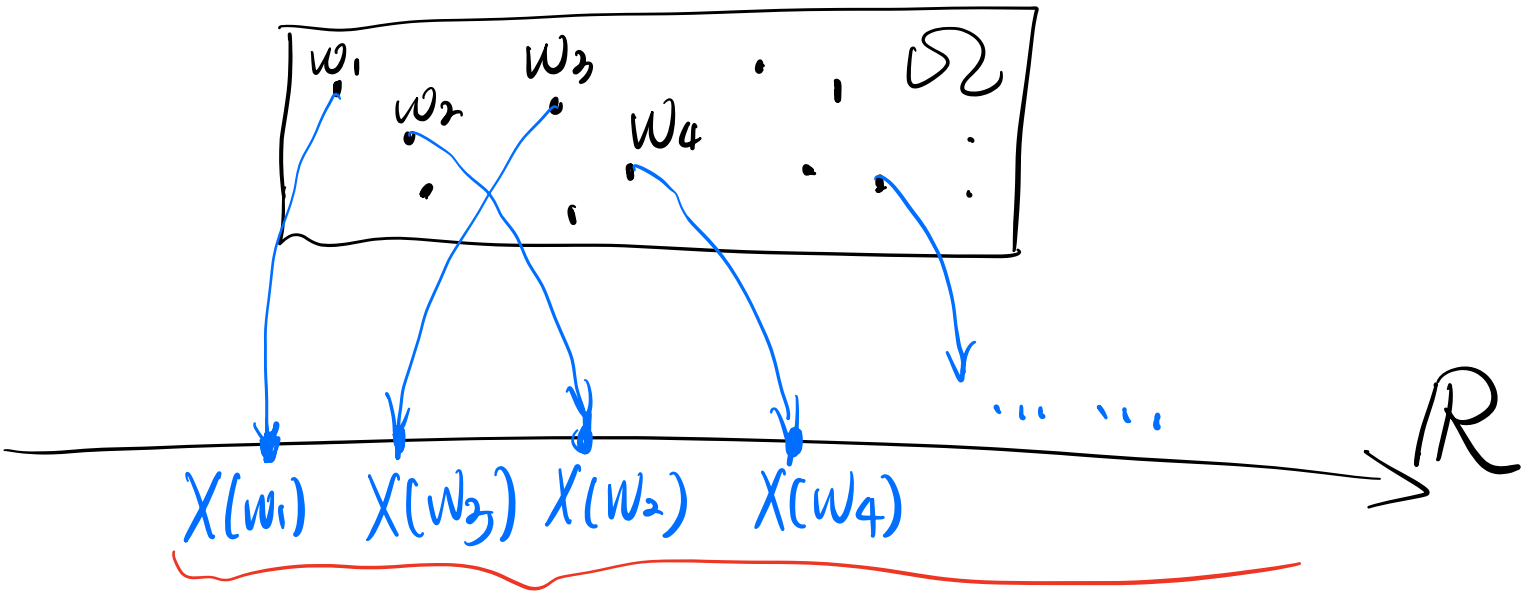
Discrete Random Variables

Definition

X is a **discrete random variable** if the range of X is a countable set, that is,

$$\{X(\omega) \mid \omega \in \Omega\}$$

is a countable set.



These numbers form a countable set.

Example. Suppose a fair coin is tossed three times. The sample space is

$$\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}.$$

Define the random variable

$$X : \Omega \to \mathbb{R}$$

by letting $X(\omega)$ be the number of heads in the outcome ω. Thus,

$$X(hhh) = 3, \quad X(hht) = 2, \quad X(hth) = 2, \quad X(thh) = 2,$$
$$X(htt) = 1, \quad X(tht) = 1, \quad X(tth) = 1, \quad X(ttt) = 0.$$

The range of X is $R_X = \{X(\omega) \mid \omega \in \Omega\} = \{0, 1, 2, 3\}$. → finite set → countable.

Since the range is finite, X is a **discrete random variable**.

Find

$$P(X = 3), \quad P(X = 2), \quad P(X = 1), \quad P(X = 0).$$

$\{X=3\} = \{hhh\}$, then $P(X=3) = \frac{1}{8}$

$\{X=2\} = \{hht, hth, thh\}$, then $P(X=2) = \frac{3}{8}$

$\{X=1\} = \{htt, tht, tth\}$, then $P(X=1) = \frac{3}{8}$

$\{X=0\} = \{ttt\}$, then $P(X=0) = \frac{1}{8}$

Note: the range of X is $\{0, 1, 2, 3\}$

and $\{X=0\}, \{X=1\}, \{X=2\}, \{X=4\}$

form a partition of the sample space Ω:

$$P(X=0) + P(X=1) + P(X=2) + P(X=3) = 1$$
$$\frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$$

Probability Mass Function

Definition

The **probability mass function** (PMF) of a discrete random variable X is defined by

$$p_X(x) = P(X = x)$$

Note:

- The notation $\{X = x\}$ represents the event

$$\{X = x\} = \{\omega \in \Omega : X(\omega) = x\},$$

consisting of all outcomes ω that are mapped to x by X.

- Hence,

$$p_X(x) = P(X = x)$$

is the probability of the event $\{X = x\}$.

- The PMF tells us how the total probability mass of 1 is distributed among the possible values of a discrete random variable.

Families of Discrete Random Variables

So far, we have constructed the PMF of a random variable directly from a particular random experiment. However, many different experiments lead to PMFs with the same mathematical form. Rather than deriving these PMFs from scratch every time, we identify and study several common families of discrete random variables.

Poisson (λ) Random Variable

A discrete random variable X is called a **Poisson random variable** with parameter $\lambda > 0$ if its probability mass function is

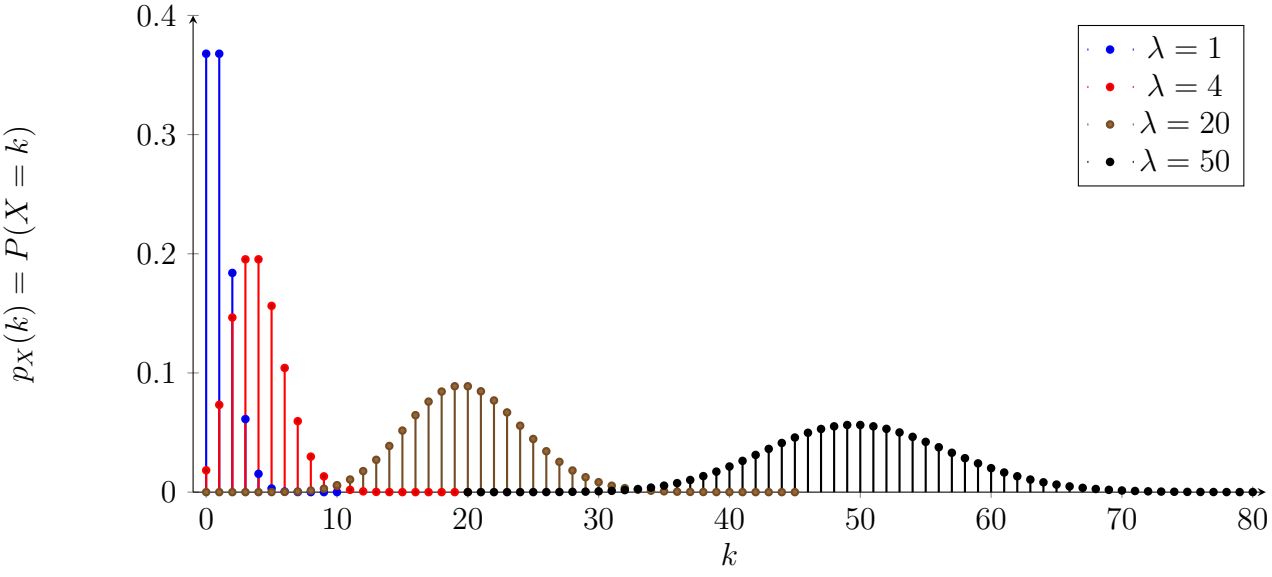
$$p_X(k) = P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

We write

$$X \sim \text{Poisson}(\lambda).$$

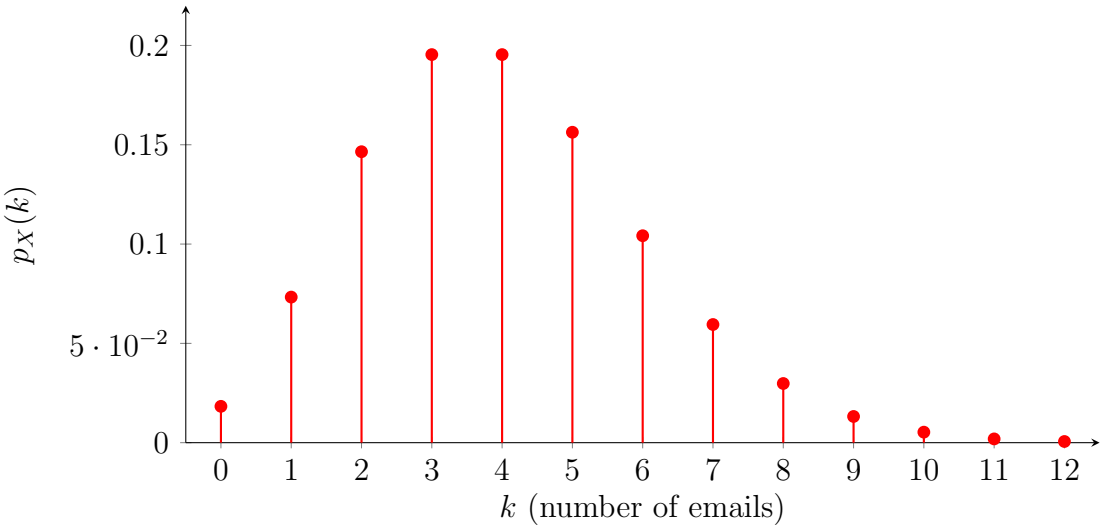
Interpretation of λ.
A Poisson random variable is commonly used to model the number of occurrences of an event in a fixed interval or region when occurrences happen randomly at a known average rate. In this setting,

$\lambda =$ average number of occurrences in that interval or region.



PMFs of Poisson random variables for different values of λ.

Notice that λ does more than appear in the formula. It tells us roughly where the probability mass is centered.



Probability mass function of $X \sim \text{Poisson}(4)$.

Example. Suppose the number of emails received during a one-hour period is modeled by a Poisson random variable with an average of 4 emails per hour. Let

$$X = \text{the number of emails received in one hour.}$$

Then

$$X \sim \text{Poisson}(4),$$

and hence

$$p_X(k) = P(X = k) = \frac{e^{-4}4^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Compare the probabilities of receiving exactly 3 and exactly 4 emails in one hour.